1. OOP Introduction

Task: Student Management System

- Ek Student class banao jo ek student ko represent kare.
- Attributes: name (string), id (integer), courses (list of strings).
- Methods:
  - add_course(course): Ek course list mein add kare.
  - remove_course(course): Ek course list se remove kare.
  - show_courses(): Student ke name, ID, aur enrolled courses print kare.
- 2 student objects banao, courses add/remove karo, aur details print karo.

Instructions:

- __init__ mein attributes initialize karo.
- courses ko list ke roop mein manage karo.
- show_courses() mein formatted output dikhayein (e.g., "Student: Ali, ID: 101, Courses: [Math, Science]").
- Error handling add karo (e.g., agar course remove karte waqt woh exist na kare).

Viva Question:

- OOP kya hai aur iske real-world applications kya hain? Encapsulation aur modularity ke faide kya hain?

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2. Classes and Objects

Task: Library Book Tracker

- Ek Book class banao jo library ki kitaabon ko track kare.
- Attributes: title (string), isbn (string), is_borrowed (boolean).
- Methods:
  - borrow_book(): Book ko borrowed mark kare agar already borrowed na ho.
  - return_book(): Book ko returned mark kare.
  - book_status(): Book ka title, ISBN, aur borrowed status print kare.
- 3 book objects banao, kuch books borrow/return karo, aur status check karo.

Instructions:

- __init__ mein is_borrowed ko default False set karo.
- borrow_book() aur return_book() mein conditions check karo (e.g., already borrowed ya not borrowed).
- book_status() mein clear output dikhayein.
- Ek edge case handle karo (e.g., already borrowed book ko dobara borrow karne ki koshish).

Viva Question:

- Class aur object mein kya farq hai? self keyword ka kya role hai?

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3. Constructors and Destructors

Task: Gadget Inventory

- Ek Gadget class banao jo electronics inventory manage kare.
- Attributes: name (string), serial_number (string), price (float).
- Constructor mein ek message print karo ke gadget added hai aur attributes set karo.
- Destructor mein ek message print karo ke gadget inventory se remove hua.
- Methods:
    - update_price(new_price): Price update kare (positive values only).
- 2 gadgets banao, price update karo, aur ek gadget delete karo.

Instructions:

- __init__ mein validation add karo ke price positive ho.
- __del__ mein serial number ke saath message print karo.
- update_price() mein invalid inputs (negative price) handle karo.
- Objects ke lifecycle (creation to deletion) test karo.

Viva Question:

- Constructor aur destructor ka purpose kya hai? __init__ aur __del__ kab call hote hain?

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4. Class Attributes vs Instance Attributes

Task: Car Dealership

- Ek Car class banao jo dealership ke cars ko manage kare.
- Class attribute: dealership_name (string, e.g., "XYZ Motors").
- Instance attributes: model (string), year (integer), price (float).
- Methods:
  - car_info(): Car ke model, year, price, aur dealership name print kare.
  - apply_discount(percentage): Price par discount apply kare.
- 2 car objects banao, ek car par discount apply karo, aur dealership name modify karke check karo ke dono cars affected hote hain ya nahi.

Instructions:

- dealership_name ko class level par set karo.
- __init__ mein instance attributes initialize karo.
- apply_discount() mein percentage ke hisaab se price kam karo aur negative price avoid karo.
- Class attribute modification ka effect test karo.

Viva Question:

- Class attributes aur instance attributes mein kya farq hai? Class attribute ko instance ke through modify karne ka kya result hota hai?

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5. Methods in Python Classes

Task: Fitness Tracker

- Ek Workout class banao jo fitness activities track kare.
- Class attribute: gym_name (string, e.g., "FitZone").
- Instance attributes: activity (string), duration (integer, minutes mein), calories (float).
- Methods:
  - Instance method: log_workout() jo activity details print kare.
  - Class method: change_gym(new_gym) jo gym name update kare.
  - Static method: is_intense(calories, duration) jo bataye ke workout intense hai ya nahi (calories/duration > 5 ho to intense).
- Ek workout object banao, details log karo, gym name change karo, aur intensity check karo.

Instructions:

- @classmethod aur @staticmethod ka sahi istemal karo.
- log_workout() mein formatted output dikhayein.
- is_intense() mein logic implement karo aur multiple test cases try karo.
- Gym name change ka effect check karo.

Viva Question:

- Instance, class, aur static methods mein kya farq hai? @classmethod aur @staticmethod kab use hote hain?

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6. Encapsulation

Task: Employee Payroll

- Ek Employee class banao jo payroll manage kare.
- Attributes:
  - Public: name (string).
  - Protected: _salary (float).
  - Private: __employee_id (string).
- Methods:
  - get_salary(): Salary return kare.
  - set_salary(amount): Salary update kare (minimum 5000 ho).
  - get_id(): Private ID ke liye "Access Restricted" return kare.
  - employee_details(): Name aur salary print kare.
- Ek employee object banao, salary update karo, aur private ID access karne ki koshish karo.

Instructions:

- _ aur __ conventions use karo.
- set_salary() mein validation add karo.
- Private attribute ko directly access karne ki koshish karo aur error note karo.
- employee_details() mein clean output dikhayein.

Viva Question:

- Encapsulation kya hai? Protected aur private attributes ko Python mein kaise implement kiya jata hai?

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7. Inheritance

Task: Online Store Products

- Ek parent class Product banao jisme attributes name (string), price (float), aur method product_info() ho jo name aur price print kare.
- Ek child class Electronics banao jo Product se inherit kare.
- Electronics mein extra attribute warranty (integer, months mein) add karo.
- product_info() ko override karke warranty bhi print karo.
- Ek Electronics object banao aur details print karo.

Instructions:

- super() use karke parent ka constructor call karo.
- product_info() mein overridden logic implement karo.
- Warranty ke liye validation add karo (e.g., positive months).
- Object banao aur method test karo.

Viva Question:

- Inheritance ka kya faida hai? super() function ka role kya hai?

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8. Polymorphism

Task: Drawing App

- Ek Tool class banao jisme ek method use_tool() ho jo "Using a tool" print kare.
- Do child classes banao: Pen aur Brush.
- Har child class mein use_tool() override karo (e.g., "Drawing with pen" aur "Painting with brush").
- Ek function apply_tool(tool) banao jo kisi bhi tool ka use_tool() method call kare.
- Dono tools ke objects banao aur function ke saath test karo.

Instructions:

- Tool ko base class banao.
- Pen aur Brush mein use_tool() override karo.
- apply_tool() function banao jo polymorphism dikhaye.
- Multiple objects ke saath function test karo.

Viva Question:

- Polymorphism kya hai? Method overriding aur duck typing mein kya farq hai?

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9. Abstraction

Task: Payment System

- Ek abstract class Payment banao using abc module.
- Isme ek abstract method process_payment(amount) ho.
- Do child classes banao: CreditCard aur PayPal.
- Har child class mein process_payment() implement karo (e.g., "Processing $50 via Credit Card" aur "Processing $50 via PayPal").
- Payment ka object banane ki koshish karo aur error note karo.

Instructions:

- ABC aur @abstractmethod ka istemal karo.
- Child classes mein method implement karo.
- Abstract class ka object banane ki koshish karo.
- Child classes ke objects banao aur payments process karo.

Viva Question:

- Abstraction ka kya matlab hai? Abstract class aur interface mein kya farq hai?

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10. The object Class

Task: Custom Data Type

- Ek Item class banao bina kisi explicit parent class ke.
- Attributes: name (string), quantity (integer).
- Method: item_info() jo name aur quantity print kare.
- Ek Item object banao aur check karo ke yeh object class ka instance hai using isinstance().
- dir() use karke object ke default methods dekho aur __str__ aur __repr__ ke output test karo.

Instructions:

- Simple class banao.
- isinstance(obj, object) test karo.
- dir() se methods explore karo.
- __str__ aur __repr__ ke default outputs print karo.

Viva Question:

- Python mein object class ka kya role hai? Har class isse kyun inherit karti hai?

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11. Special (Magic/Dunder) Methods

Task: Shopping Cart

- Ek Cart class banao jo shopping cart manage kare.
- Attributes: items (dictionary with item names as keys and quantities as values).
- Special methods:
  - __setitem__(name, quantity): Item aur quantity add kare.
  - __getitem__(name): Item ka quantity return kare ya "Item not found" dikhaye.
  - __len__: Total items (unique names) count kare.
  - __add__(other): Do carts ke items combine kare aur naya cart return kare.
- Ek cart banao, items add karo, quantity check karo, aur do carts combine karo.

Instructions:

- items dictionary ko __init__ mein initialize karo.
- Special methods implement karo.
- __add__ mein new cart object return karo.
- Combined cart ke contents test karo.

Viva Question:

- Special methods ka kya kaam hai? __setitem__ aur __getitem__ ka use kya hai?

Viva Preparation Tips

1. **Code Practice:** Har task ke liye code likhein aur output verify karein. Edge cases (e.g., invalid inputs, empty lists) test karein.
2. **Concept Clarity:** Har task ke baad related viva question ka jawab likhein ya verbally practice karein.
3. **Debugging:** Agar code mein error aaye, to uska root cause samjhein aur fix karein. Viva mein error handling ke sawal common hote hain.
4. **Explanation:** Har task ke code ko explain karne ki practice karein, jaise aap viva mein teacher ko samjha rahe hain.
5. **Time Management:** Har task ko 15-20 minutes mein complete karne ki koshish karein taake aap efficient ho jayein.

Kaise Tayyari Karein?

- Ek notebook ya IDE mein har task ke liye alag section banao.
- Task complete karne ke baad output screenshot ya notes save karo for revision.
- Viva questions ke jawab likh kar memorize karein ya kisi friend ke saath mock viva karein.
- Agar koi task ya concept mushkil lage, to mujhe batayein, mein aur simplify karunga.

Yeh tasks aapko **intermediate-level practice** aur **viva confidence** denge. Start coding and let me know if you need help with any task! **Best of luck!** 🚀